Space Syntax: comparing six towns

Mean Depth · RA · RRA · Integration HH · Choice · Choice norm

Modeling urban space through space syntax approaches

Student: SOUKOU A. Ezechiel (Master 2 GMS)

Professor: Dr Thomas Gloaguen

La Carolina | Keerbergen | Cumbernauld | Nördlingen | Suloszowa | Saint-Nectaire

What each index measures

Read from DepthmapX attribute tables (averages over all segments of each map)

1 Mean Depth

Average number of steps from one segment to all others. Low = shallow, close to everything.

2 RA

Mean Depth rescaled to 0–1: 2(MD−1)/(n−2). Still depends on the size of the town.

3 RRA

RA divided by D_n (a reference value for a system of the same size). Removes the size effect: the index to compare towns. Low = integrated.

4 Integration HH

1/RRA, the same information on a reversed scale. High = integrated.

5 Choice

Number of shortest paths passing through a segment (through-movement). Grows with size, so raw values are not comparable between towns.

6 Choice norm

Choice rescaled for comparison. Its mean equals the mean RA, so use its max and its spread.

All six towns at a glance

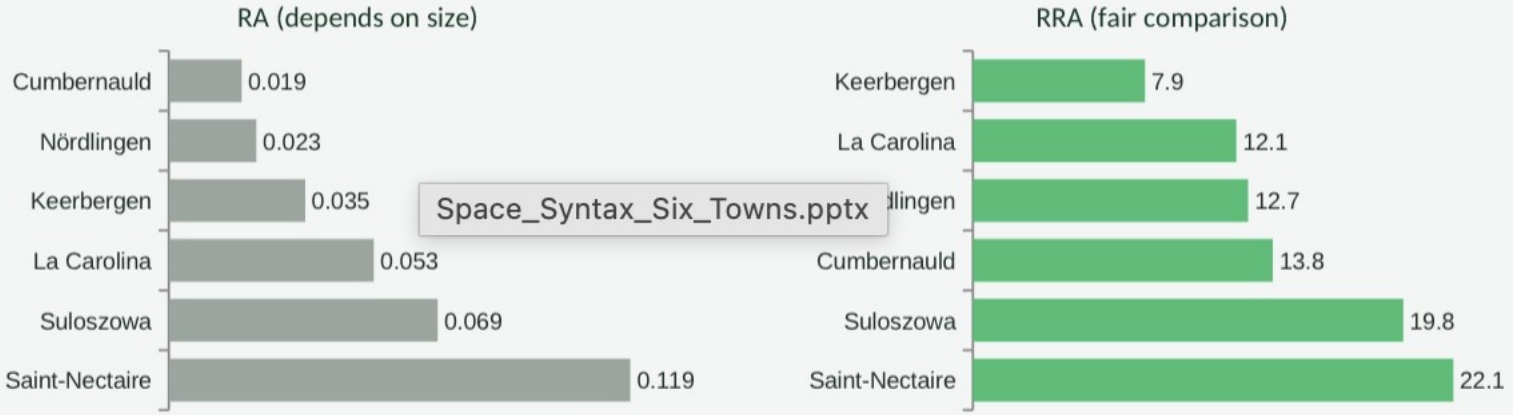
Sorted from most to least integrated (RRA)

Town	n	Mean Depth	RA	RRA	Int. HH	Choice avg	Choice max	Choice norm max
Keerbergen	4,271	76.2	0.0352	7.93	0.133	3.2e5	5.0e6	0.551
La Carolina	4,368	116.4	0.0529	12.14	0.096	5.0e5	4.2e6	0.444
Nördlingen	12,384	140.6	0.0226	12.68	0.083	1.7e6	4.9e7	0.638
Cumbernauld	16,880	159.3	0.0188	13.82	0.076	2.7e6	1.0e8	0.734
Suloszowa	5,650	196.9	0.0694	19.83	0.053	1.1e6	1.0e7	0.647
Saint-Nectaire	3,405	203.6	0.1191	22.14	0.047	6.9e5	5.7e6	0.993

Town size ranges from 3,405 to 16,880 segments: raw Choice, Mean Depth and RA cannot be compared directly, RRA and Choice norm can.

Integration: size hides the truth in RA

The same six towns ranked by RA and by RRA

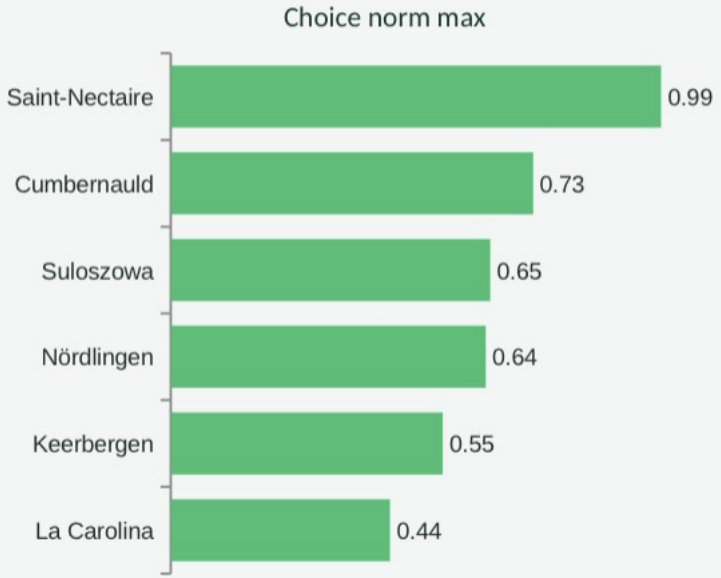


Space_Syntax_Six_Towns.pptx

- By RA, Cumbernauld looks the most integrated only because it is the largest town (16,880 segments); by RRA it falls to 4th.
- Keerbergen is clearly the most integrated; its worst segment (14.9) is barely worse than Cumbernauld's average (13.8). Saint-Nectaire's best segment (15.1) is worse than the average of four towns.

Choice: who concentrates movement?

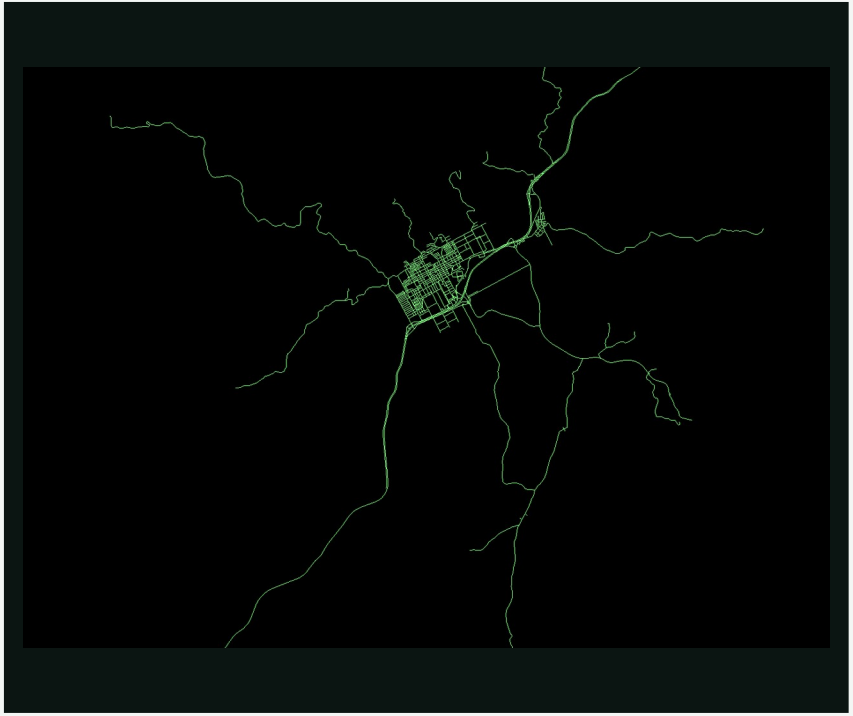
Choice norm, maximum value per town



- Raw Choice max: Cumbernauld 1.0e8 vs La Carolina 4.2e6. Mostly a size effect, so not comparable.
- Saint-Nectaire (0.99): one route carries almost every shortest path.
- Cumbernauld and Nördlingen: the most unequal (SD/mean 3.1 and 2.9): few streets carry the movement, most carry none.
- La Carolina: lowest max and flattest distribution (SD/mean 1.4): many alternative routes, typical of a grid.

La Carolina

Compact grid core, long rural tentacles



12.1	0.44	4,368
RRA · rank 2/6	Choice norm max	segments

- Map: a dense orthogonal grid in the centre; outside it only a few long, winding roads radiate out, with almost no cross-links.
- Integration: the grid is highly integrated, but every long road adds deep segments. RRA reaches 36.8 (mean 12.1) and its spread is twice that of the other towns: the strongest core–periphery contrast.
- Choice: the lowest max (0.44). A grid offers many equivalent routes, so through-movement is spread out; peaks are expected on the roads linking the grid to the outside.

Keerbergen

Evenly meshed suburban network

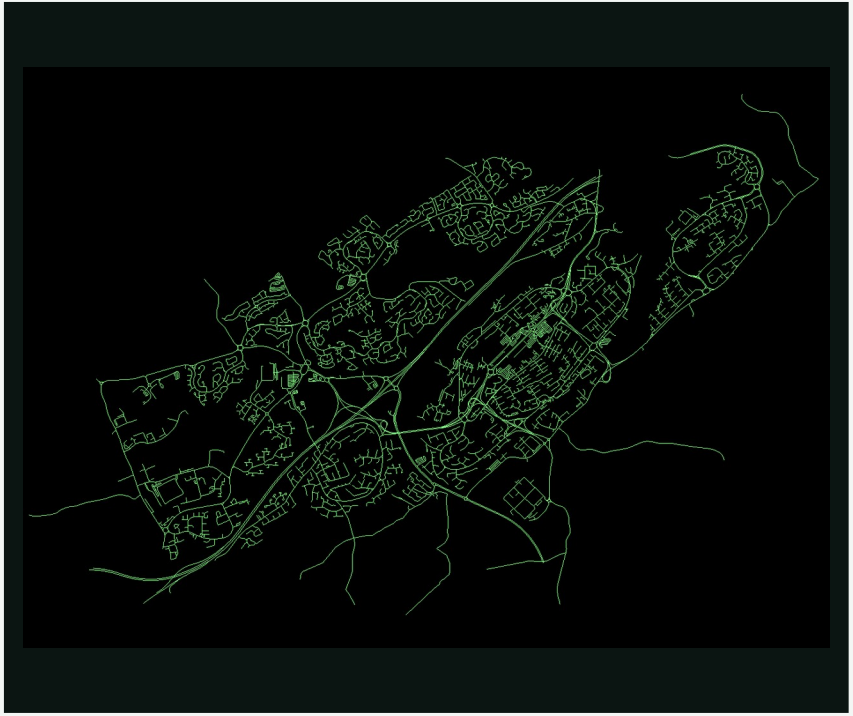


7.9	0.55	4,271
RRA · rank 1/6	Choice norm max	segments

- Map: a continuous, irregular mesh of connected streets covering the whole area, with hardly any isolated tentacles.
- Integration: the shallowest town (Mean Depth 76). Even its least integrated segment (RRA 14.9) is close to Cumbernauld's average, and the low spread (SD 1.9) shows a homogeneous network.
- Choice: moderate max (0.55): a few diagonal through-roads stand out, but alternatives exist.
- Caveat: the map is cropped tightly around the built area, which lowers depth.

Cumbernauld

Motorway spine, tree-like neighbourhoods

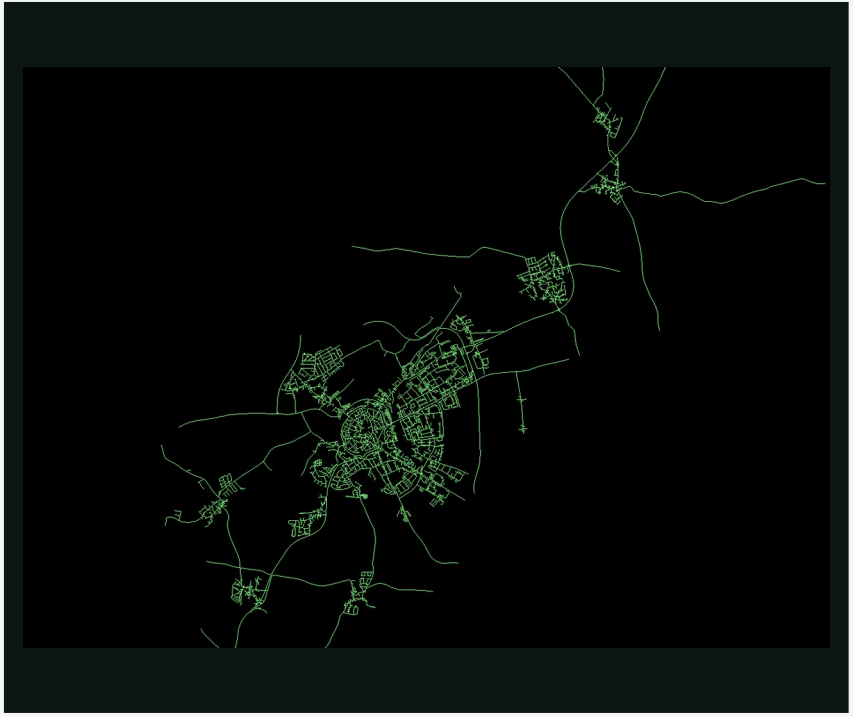


13.8	0.73	16,880
RRA · rank 4/6	Choice norm max	segments

- Map: a diagonal high-speed road cuts through the town; neighbourhoods are loops and cul-de-sacs hanging off a few distributors; the centre is a dense cluster on the east side.
- Integration: only 4th despite its size. The hierarchical layout adds depth; RA ranks it first only because n = 16,880.
- Choice: highest raw values (max 1.0e8) and the most unequal distribution (SD/mean 3.1): a few distributors and the spine carry through-movement, most streets almost none.

Nördlingen

Circular core with satellite villages

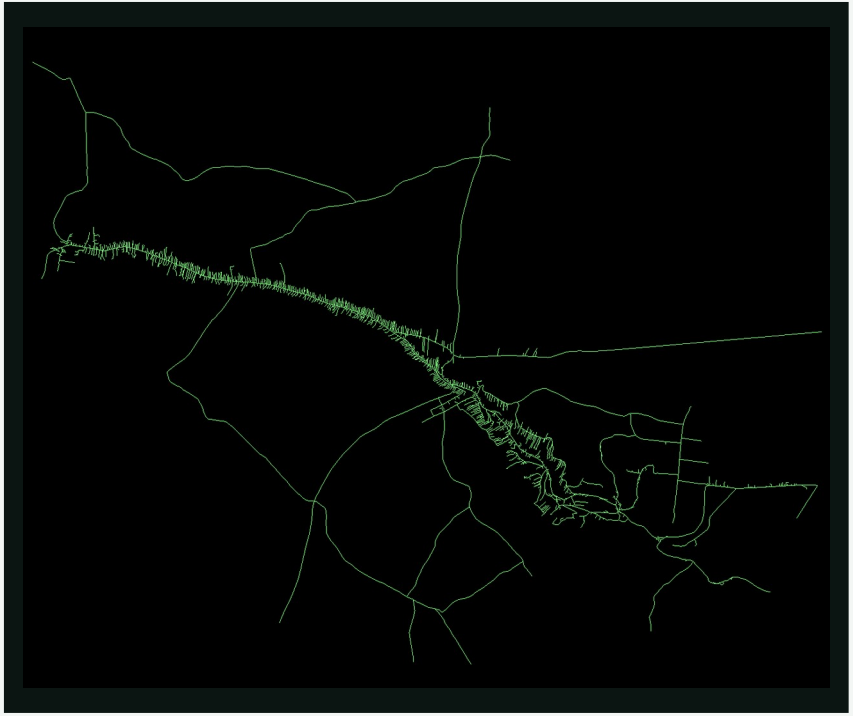


12.7	0.64	12,384
RRA · rank 3/6	Choice norm max	segments

- Map: a dense concentric core (ring streets and radials), surrounded by small villages strung along long roads.
- Integration: 3rd, very close to La Carolina. The core is shallow; villages hang off single roads and are deeper (RRA max 26.7).
- Choice: second most unequal (SD/mean 2.9). The ring and the radial roads linking villages to the core are bottlenecks: every village–core trip must use them.

Suloszowa

Ribbon village along one road

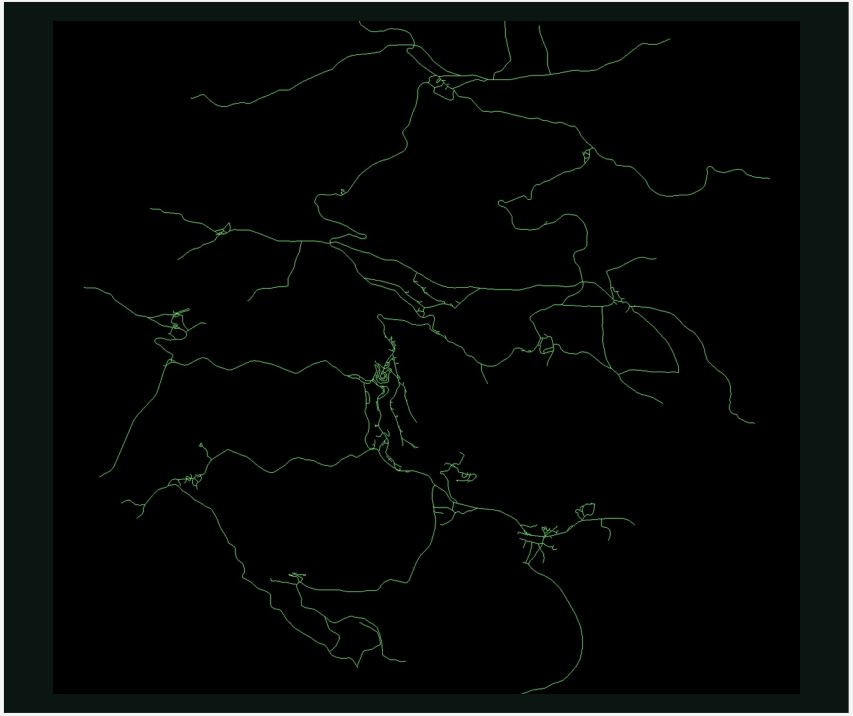


19.8	0.65	5,650
RRA · rank 5/6	Choice norm max	segments

- Map: a very long main road lined with hundreds of short perpendicular spurs (plot access), plus a few rural connectors.
- Integration: 5th. The spurs are dead ends, and going from one end of the ribbon to the other takes many steps.
- Choice: the main road is the spine, but rural loops offer some alternatives, so it is less extreme than Saint-Nectaire.

Saint-Nectaire

Sparse clusters on long rural roads



22.1	0.99	3,405
RRA · rank 6/6	Choice norm max	segments

- Map: a few tiny clusters far apart, linked by long winding roads; almost no meshing.
- Integration: the least integrated town (Mean Depth 204). Its best segment (RRA 15.1) is worse than the average of four other towns. With only 3,405 segments, the cause is distance, not size.
- Choice: max 0.99. A single route lies on almost every shortest path: a tree-like network with one dominant link; other segments score near zero.

What the comparison tells us

1

Compare with RRA

RA and raw Choice mostly reflect town size. RRA / Integration HH and Choice norm are the fair indices.

2

Form drives integration

Meshed grids and networks (Keerbergen, La Carolina, Nördlingen core) are shallow; hierarchical layouts (Cumbernauld) and linear or sparse ones (Suloszowa, Saint-Nectaire) are deep.

3

Choice reveals structure

Grids spread movement over many routes; tree-like or hierarchical networks concentrate it on a few links.

Integration ranking (RRA): Keerbergen › La Carolina › Nördlingen › Cumbernauld › Suloszowa › Saint-Nectaire

Limit: results also depend on map extent and analysis radius (tight crops lower depth).